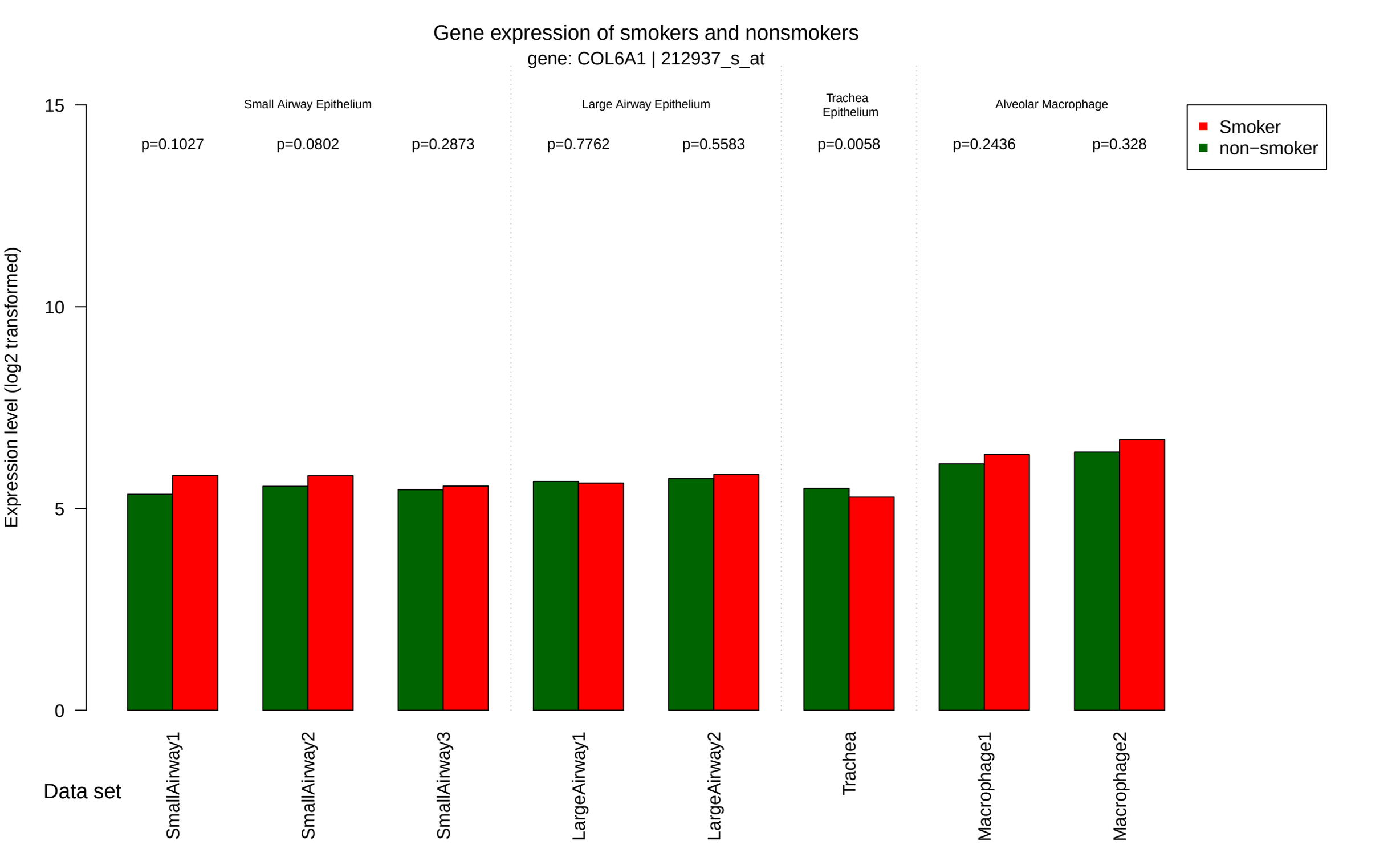
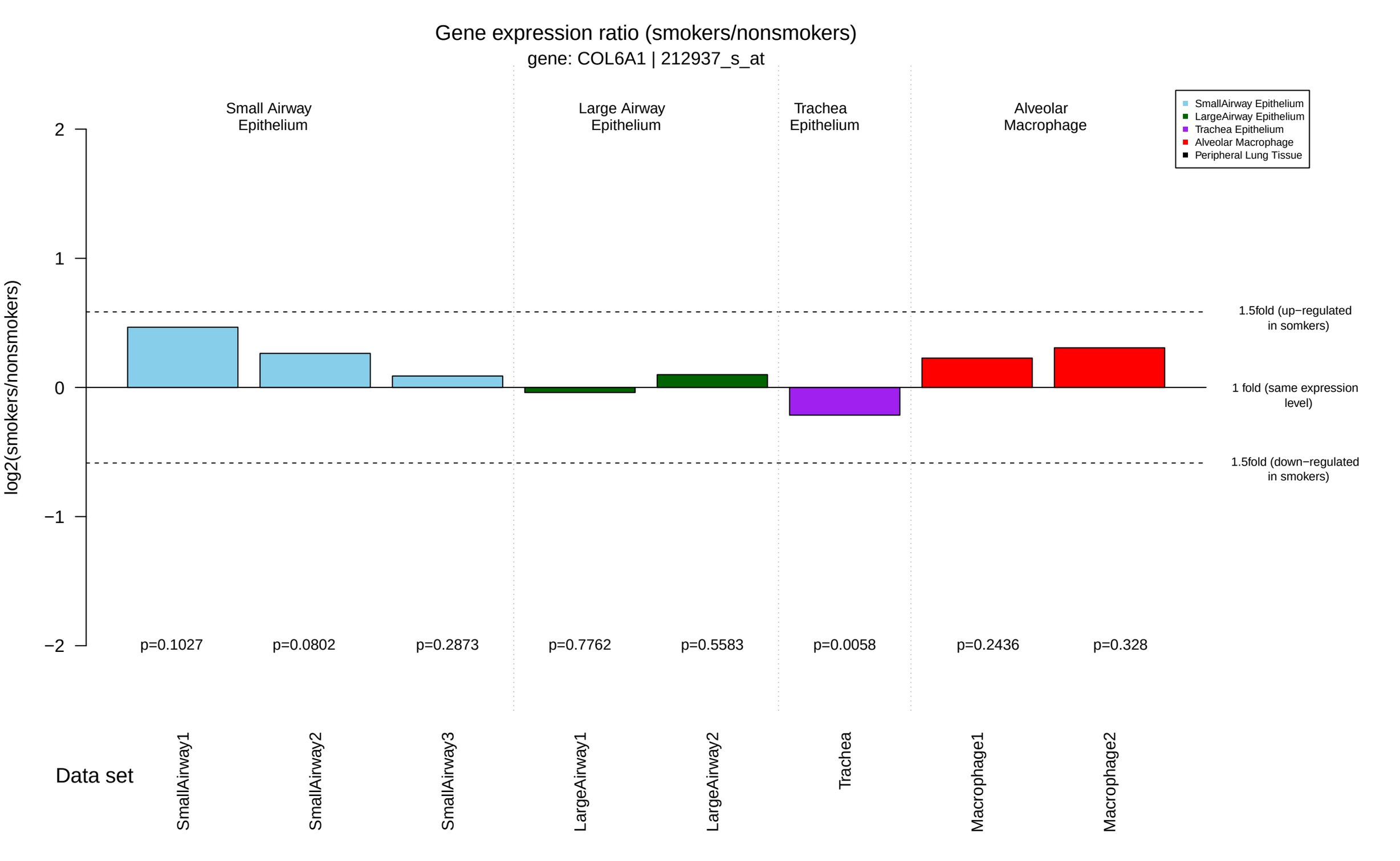
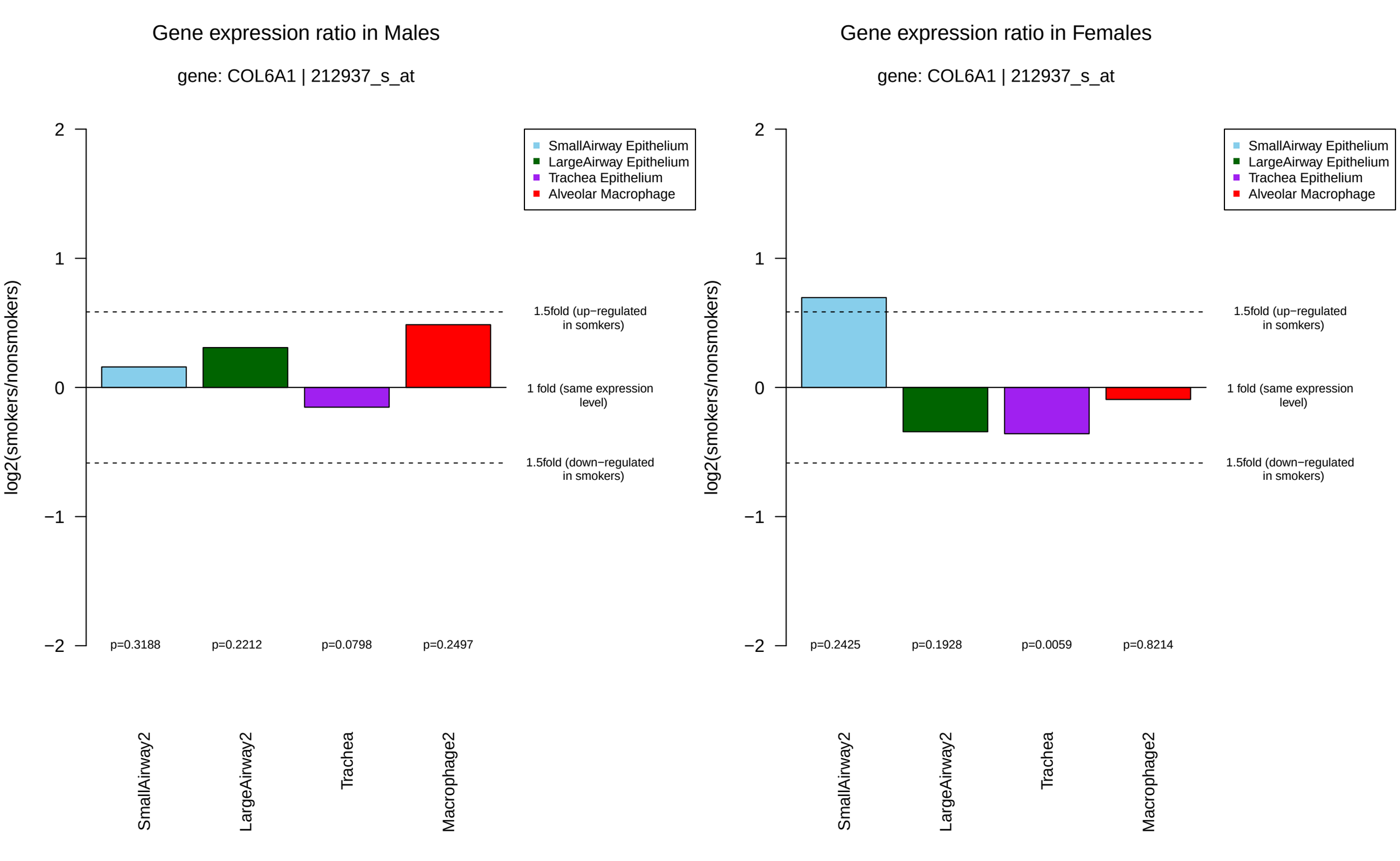
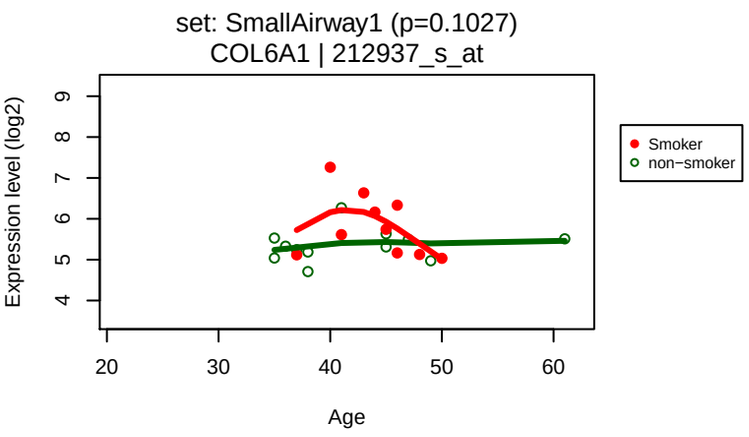
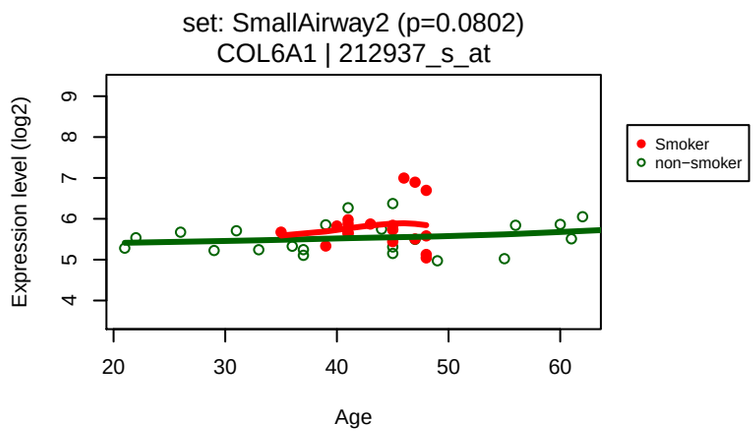
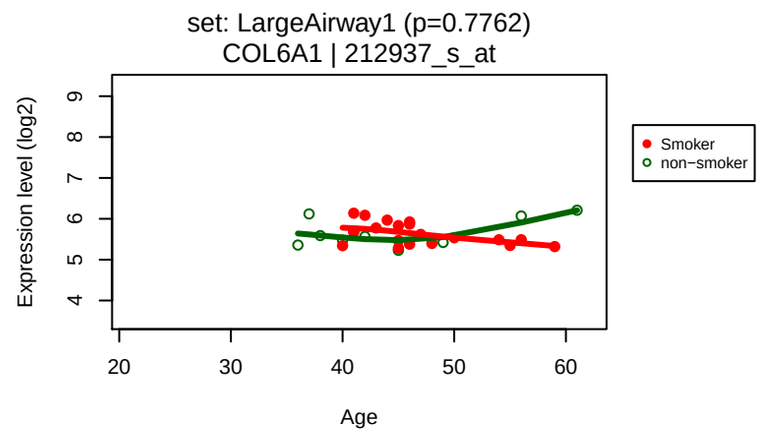

Gene expression of smokers and nonsmokers
gene: COL6A1 | 212937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1027
p=0.0802
p=0.2873
p=0.7762
p=0.5583
p=0.0058
p=0.2436
p=0.328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

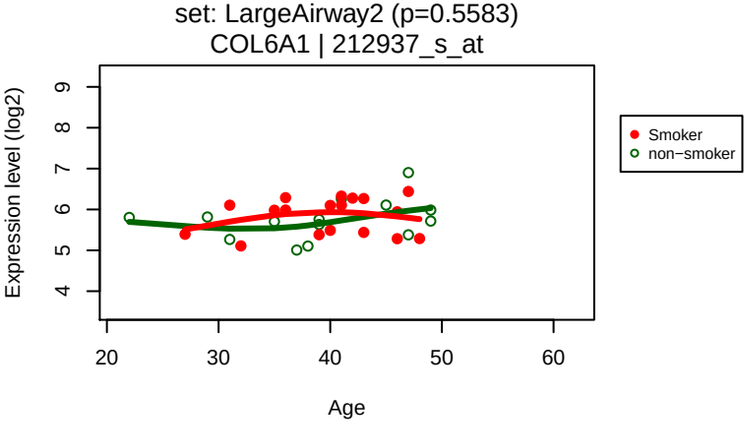

Gene expression ratio (smokers/nonsmokers)
gene: COL6A1 | 212937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1027
p=0.0802
p=0.2873
p=0.7762
p=0.5583
p=0.0058
p=0.2436
p=0.328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

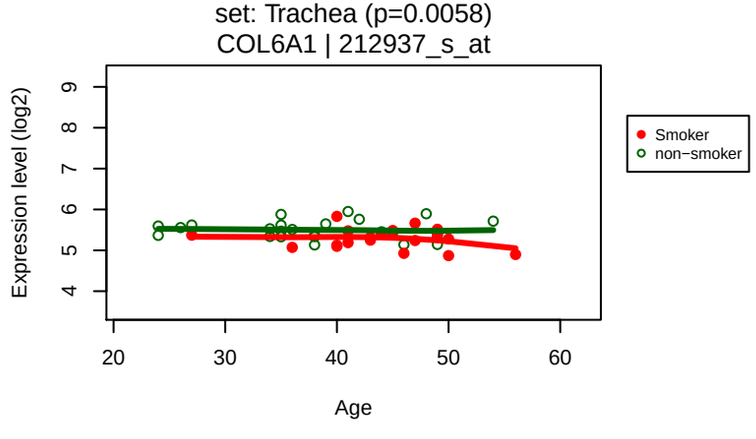

Gene expression ratio in Males
gene: COL6A1 | 212937_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3188
p=0.2212
p=0.0798
p=0.2497
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL6A1 | 212937_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2425
p=0.1928
p=0.0059
p=0.8214
SmallAirway2
LargeAirway2
Trachea
Macrophage2

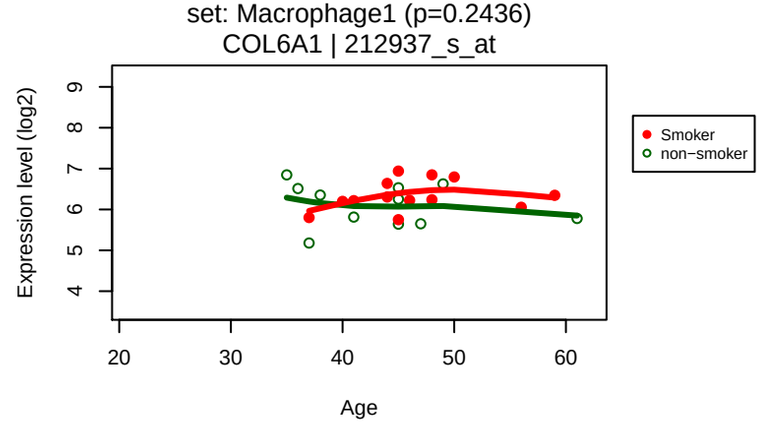

set: SmallAirway1 (p=0.1027)
COL6A1 | 212937_s_at
Expression level (log2)
Age
Smoker
non−smoker

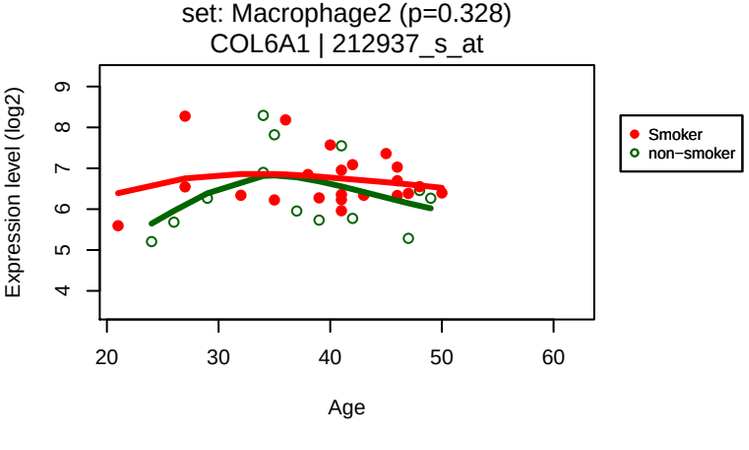

set: SmallAirway2 (p=0.0802)
COL6A1 | 212937_s_at
Expression level (log2)
Age
Smoker
non−smoker

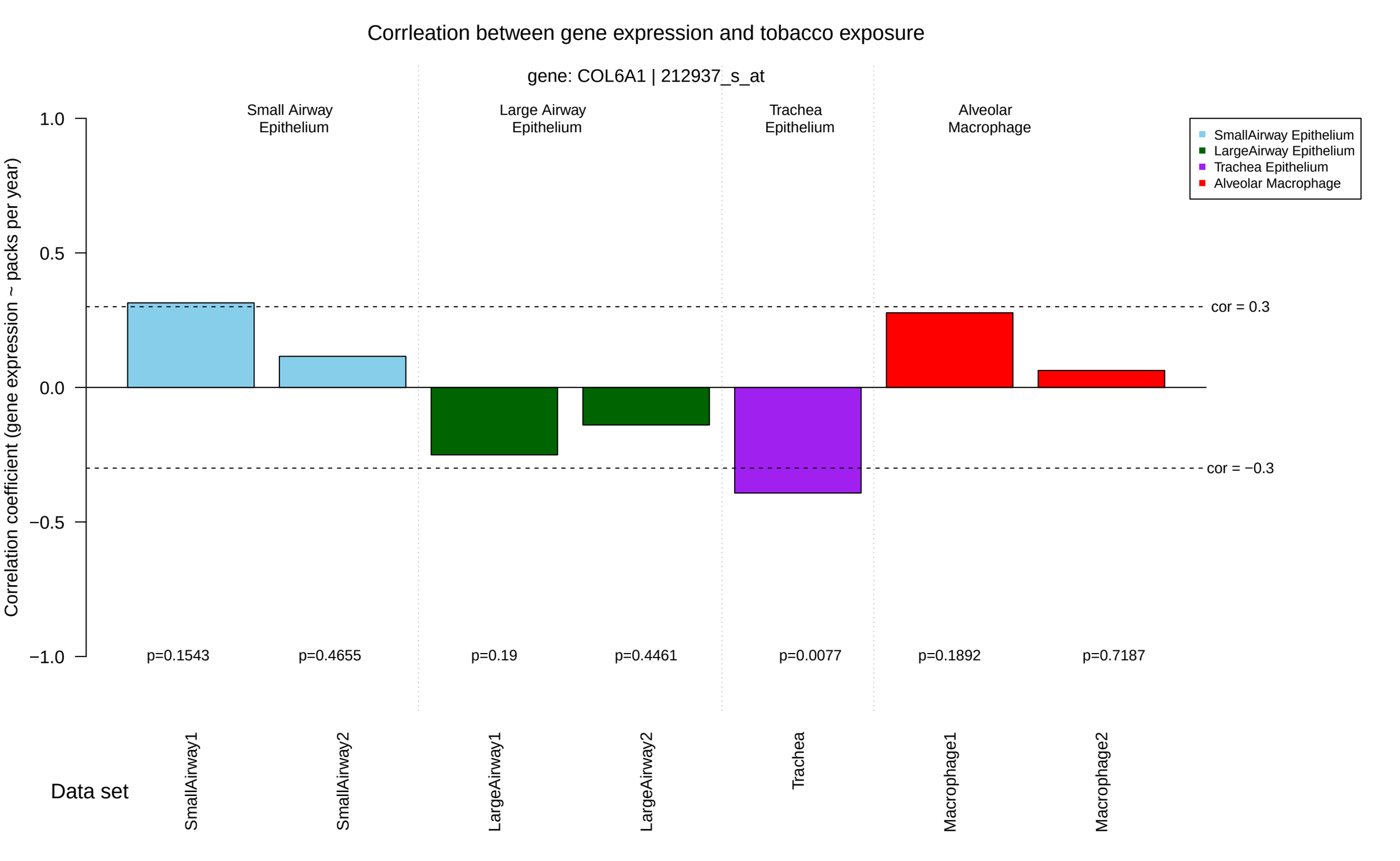

set: LargeAirway1 (p=0.7762)
COL6A1 | 212937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5583)
COL6A1 | 212937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0058)
COL6A1 | 212937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2436)
COL6A1 | 212937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.328)
COL6A1 | 212937_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL6A1 | 212937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1543
p=0.4655
p=0.19
p=0.4461
p=0.0077
p=0.1892
p=0.7187
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2